FEASIFY INTEGRATION & END-TO-END TEST REPORT

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1. Executive Summary

This report documents the verification of end-to-end integration workflows connecting the Feasify frontend React application, Node.js/Express backend services, Cloud Firestore database, and external interfaces (OpenStreetMap Nominatim, Gemini AI cascade, and DTI/SEC verification fallbacks).

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2. End-to-End Workflow Scenarios

Scenario 1: Student Proposal Creation to Adviser Review

```
sequenceDiagram
    autonumber
    actor Student as Student Team Leader
    participant Projects as Projects.tsx
    participant Firestore as Cloud Firestore
    participant Socket as Socket.io Backend
    actor Adviser as Faculty Adviser
    participant AdvDash as AdviserDashboard.tsx

    Student->>Projects: Fill Proposal Details (Title, Location, Products)
    Projects->>Firestore: addDoc("proposals", { status: "Draft" })
    Firestore-->>Projects: Doc ID generated (prop-01)
    Student->>Projects: Click "Submit Proposal"
    Projects->>Firestore: updateDoc("proposals/prop-01", { status: "Submitted" })
    Projects->>Firestore: updateDoc("groups/grp-01", { status: "Pending Review" })
    Projects->>Socket: emit("proposal_submitted", { groupId, proposalId })
    Socket-->>Adviser: Real-time broadcast / notification badge increment
    Adviser->>AdvDash: Navigate to assigned section
    AdvDash->>Firestore: query("proposals", where("groupId", "==", "grp-01"))
    Firestore-->>AdvDash: Return prop-01 with status "Submitted"
    Note over AdvDash: Adviser reviews products, financial projections, and AI score
```

- **Execution Outcome: Passed.** The proposal was created as a draft, submitted, persisted in Firestore, and successfully appeared in the Adviser Dashboard for the matching section.

Scenario 2: Proposal Limit Enforcement (Strict 3-Proposal Ceiling)

```
flowchart TD
    Start["Student Initiates Proposal Creation"] ⟶ Step1["Count Check: proposals.length in React
    Step1 ⟶|Count < 3| Step2["Network Query: GET /api/teams/:id/proposals/count"]
    Step2 ⟶|Count < 3 & allowed=true| Allow["Allow Proposal Creation & Form Input"]
    Step1 ⟶|Count ≥ 3| Block1["Client State Block: Show Toast (Limit Reached)"]
    Step2 ⟶|Count ≥ 3 & allowed=false| Block2["Backend Count Block: Show Quota Modal"]
    Allow ⟶ Attempt4["Adversary sends POST /api/teams/:id/proposals at count=3"]
    Attempt4 ⟶ Block3["Backend API Rejection: HTTP 400 (proposalLimitReached: true)"]
```

- **Test Case Steps:**
  1. Student group created Proposal 1 (Count = 1).
  2. Student group created Proposal 2 (Count = 2).
  3. Student group created Proposal 3 (Count = 3).
  4. Student attempted to create a 4th proposal via UI: Blocked by `proposals.length ≥ 3` check; toast displayed.
  5. Student refreshed the browser: State re-hydrated from Firestore; creation button remained disabled.
  6. Backend endpoint tested: `POST /api/teams/:id/proposals` rejected with HTTP 400.
- **Execution Outcome: Passed with Vulnerability Note.** While both frontend and backend APIs enforce the limit, direct invocation of `addDoc` via client-side Firestore SDK can bypass the limit because of absent database-level quota rules.

Scenario 3: Adviser Feedback, Revision Request, and Resubmission

```
sequenceDiagram
    autonumber
    actor Adviser as Faculty Adviser
    participant AdvDash as AdviserDashboard.tsx
    participant Firestore as Cloud Firestore
    actor Student as Student Leader
    participant Projects as Projects.tsx

    Adviser⟶⟶AdvDash: Input Remarks: "Adjust markup on Product B to 45%"
    Adviser⟶⟶AdvDash: Select Action "Request Revision"
    AdvDash⟶⟶Firestore: updateDoc("proposals/prop-01", { status: "Revision Required", adviserRemar
    AdvDash⟶⟶Firestore: addDoc("notifications", { recipientId: studentId, title: "Revision Request
    Student⟶⟶Projects: Open Proposal Management
    Projects⟶⟶Firestore: onSnapshot("proposals")
    Firestore⟶⟶Projects: Updated proposal with "Revision Required" badge and remarks
    Student⟶⟶Projects: Edit Product Costing in Financial Input
    Student⟶⟶Projects: Click "Submit Revised Proposal"
    Projects⟶⟶Firestore: updateDoc("proposals/prop-01", { status: "Submitted" })
```

- **Execution Outcome: Passed.** Real-time snapshots propagate status changes between Adviser and Student dashboards seamlessly without full page refreshes.

Scenario 4: Chairperson Oversight and Multi-Role Consistency

- **Workflow:**
  1. Student submitted proposal in Section `BSBA-FM-4A`.
  2. Adviser evaluated and recommended proposal in Adviser Dashboard.
  3. Chairperson accessed `/admin/projects` (`ChairpersonFeasib.tsx`).
  4. Filtered by Section `BSBA-FM-4A`.
- **Execution Outcome: Passed.** The Chairperson module reflects the identical proposal data, financial inputs, and adviser recommendation history, confirming consistent data models across all three authenticated roles.

Scenario 5: Business Name & Trademark Collision Checking

- **Workflow:**
  1. Student typed "Cocopan Fresh Bakery" into Business Name field.
  2. Frontend invoked `checkName("Cocopan Fresh Bakery", "all")`.
  3. Backend `NameCheckerService` detected trademark collision against well-known registered brand "cocopan".
  4. Response returned: `trademarkAlert: { isProtected: true, matchedBrand: "cocopan", type: "well_known" }`.
  5. Frontend UI rendered amber warning badge alerting the student of potential trademark conflict.
- **Execution Outcome: Passed.** Accurate detection of trademark collisions prevents students from submitting infringing business names for capstone defense.

Scenario 6: Location Selection and Coordinate Persistence

- **Workflow:**
  1. Student clicked "Pick on Map" in proposal form.
  2. Leaflet initialized OpenStreetMap container.
  3. Student searched "Pamantasan ng Lungsod ng Valenzuela".
  4. Autocomplete suggestion selected; pin moved to latitude `14.6991`, longitude `120.9754`.
  5. Nominatim reverse-geocoded coordinates into: `"Poblacion II, Malinta, Valenzuela, Metro Manila"`.
  6. Proposal saved and closed.
  7. Proposal reopened: Form displayed the clean formatted address string.
- **Execution Outcome: Passed.** Clean human-readable address stored without exposing raw coordinate floats.

Scenario 7: Unauthorized Access & Route Protection

- **Workflow:**
  1. Unauthenticated client attempted direct HTTP GET to `/dashboard` and `/projects`.
  2. `ProtectedRoute.tsx` intercepted unauthenticated state and redirected to `/`.
  3. Authenticated Student attempted to access `/admin/users`.
  4. `ProtectedRoute.tsx` inspected `authState.role` and redirected Student back to `/dashboard`.
- **Execution Outcome: Passed.** Frontend client route protection operates as designed.

Scenario 8: Network Interruption Resilience & Fallback Handling

- **Workflow:**
  1. Simulated backend companion server offline (`http://localhost:10000` unreachable).
  2. Student attempted proposal submission in `Projects.tsx`.
  3. `Projects.tsx` caught network failure on `GET /api/teams/:id/proposals/count` and gracefully fell back to client-side Firestore state verification (`proposals.length ≥ 3`).
  4. Proposal saved directly to Firestore.
- **Execution Outcome: Passed.** Client application demonstrates high resilience against backend companion outages.
